

Write logic for this:

1. A retail chain wants to forecast product sales using factors like holidays, promotions, location, and inventory. The dataset has 80 features. How should they select the most impactful ones?

Answer:

- Analyze and preprocess the dataset and separate the input features from the sales target.
 - After splitting the data into training and testing sets, use Random Forest feature importance to identify the most impactful features among the 80 features.
 - Then I would select the important features, train a regression model using them, make predictions on the test data.
 - Evaluate the model using metrics such as MAE, RMSE, and R^2 .
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2. A podcast platform wants to recommend new podcasts to users based on the categories and hosts they follow. How should they design this content-based system?

Answer:

- Use content-based filtering. Analyze the categories and hosts of the podcasts the user follows or listens to.
 - Create a user preference profile and represent each podcast using its categories and host information.
 - Calculate the similarity between the user profile and available podcasts, and recommend the most similar podcasts.
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3. An industrial firm wants to predict machine failure using sensor readings and system logs (200+ features). How do they choose which inputs to use?

Answer:

- Analyze the dataset and separate input features (X) and target (Machine Failure).
- Preprocess the data - handle missing values, outliers, and encode categorical data if needed.
- Split the data into training and testing sets.
- Since there are 200+ features, apply feature selection to identify the most useful inputs.
- Use methods such as:
 - **SelectKBest** → selects the top K statistically relevant features.

- **RFE** → recursively removes less important features.
 - **Random Forest Feature Importance** → identifies features that contribute most to prediction.
 - Train the model using the selected features.
 - Test and evaluate the model using appropriate classification metrics such as Precision, Recall, F1-score, and ROC-AUC.
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4. A fashion app wants to suggest outfits to users based on their previous likes, color preferences, and seasonal trends. How should they implement the recommendation?

Answer:

- Collect user preferences - previous likes and preferred colors.
 - Analyze the user's past interactions to understand their preferences.
 - Analyze outfit/item features such as color, style, category, and season.
 - Use content-based filtering to match the user's preferences with similar outfits.
 - Include seasonal trends as an additional feature.
 - Recommend the most relevant outfits to the user.
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5. A research lab wants to predict heart disease using patient data. The dataset has lab results, habits, and vitals. How should they select only the most relevant features?

Answer:

- Analyze the dataset and separate input features (X) and target (Y = heart disease).
 - Preprocess the data:
 - Handle missing values using `fillna()` / `dropna()`
 - Encode categorical features if needed
 - Apply scaling if the selected algorithm requires it.
 - Split the data into training and testing sets.
 - Apply feature-selection techniques such as:
 - SelectKBest → selects statistically relevant features.
 - RFE → recursively removes less important features.
 - Random Forest Feature Importance → identifies important features based on their contribution.
 - Select the most relevant features.
 - Train the classification model using those selected features.
 - Test and evaluate the model using metrics such as Precision, Recall, F1-score, and ROC-AUC
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6. A job portal wants to recommend job listings based on a user's resume and previous applications. How can they build a recommendation system?

Answer:

- Analyze the user's resume - skills, education, experience, job roles, etc.
 - Analyze previous applications - jobs the user applied for or showed interest in.
 - Use content-based filtering to find jobs whose requirements match the user's resume and previous interests.
 - Use collaborative filtering to find jobs that users with similar application behavior liked or applied for.
 - Combine both approaches to create a hybrid recommendation system.
 - Recommend the most relevant jobs to the user.
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7. A smart home system wants to predict energy usage using readings from multiple sensors. With 100+ features, how do they reduce complexity?

Answer:

- The dataset has more than 100 features, reduce the complexity using feature-selection or dimensionality-reduction techniques.
 - Such as SelectKBest, RFE, Random Forest Feature Importance, or PCA.
 - After selecting or transforming the features, train a regression model and evaluate its performance using MAE, RMSE, and R^2
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8. An e-learning platform wants to suggest tutorials based on a learner's skill gaps identified from quizzes. How should they build this system?

Answer:

- Analyze the learner's quiz results to identify their skill gaps.
 - Analyze the skills covered by each tutorial and use content-based filtering to match the learner's weak areas with relevant tutorials.
 - Finally, recommend tutorials that help improve those skills and update the recommendations based on future quiz performance.
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9. A social media app wants to predict post engagement (likes/comments) using post features, timing, and user activity. How do they select the right features?

Answer:

- Preprocess the post, timing, and user-activity data and separate, the target engagement variable. After splitting the data into training and testing sets,
 - Use SelectKBest, RFE, or Random Forest Feature Importance to identify the most relevant features.
 - Train a regression model using the selected features and evaluate it using MAE, RMSE, and R^2 .
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10. Now you want to create an app named attendance in the project and register it.write logic for this Django project

Answer:

- Create the Django project using `startproject`.
 - Create the attendance application using `python manage.py startapp attendance`.
 - register the app by adding 'attendance' to `INSTALLED_APPS` in `settings.py`.
 - Run migrations if models are created and start the development server.
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